02. PIT and other transforms

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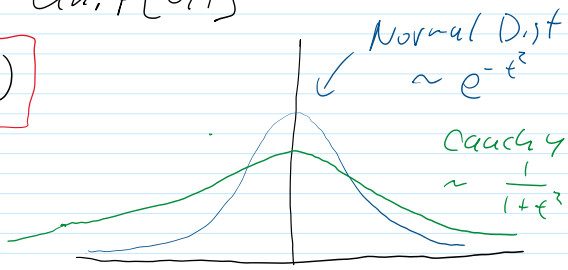
PIT: • we want to simulated / generate iid random variables $Z_1, \dots, Z_n$ which have CDF $F(t)$

• We have $U_1, \dots, U_n \overset{iid}{\sim} Unif[0,1]$

• Solution: $\boxed{Z_i := F^{-1}(U_i)}$



Cauchy Distribution

PDF: $f(t) = \frac{1}{\pi(1+t^2)}$

CDF: $F(t) = \frac{1}{\pi}\arctan(t) + \frac{1}{2} \in [0,1]$

PIT: $U = F(Z)$ ← solve this for Z

$U = \frac{1}{\pi}\arctan(Z) + \frac{1}{2}$

$\pi(U - \frac{1}{2}) = \arctan(Z)$

$\boxed{\tan[\pi(U - \frac{1}{2})] = Z}$

This is exactly what R does when you use rcauchy()

t-distribution:

comes from the Normal Dist but when we estimate the unknown variance.

Let $Z \sim N(0,1)$, $V \sim \chi^2(\nu)$, for $\nu \in \mathbb{N}$ or $\nu > 0$

Also, Z, V are independent

Then $T = \frac{Z}{\sqrt{V/\nu}} \sim t(\nu)$

* If we want to generate a $T \sim t(\nu)$ we could generate $Z \sim N(0,1)$ and $V \sim \chi^2(\nu)$ and compute $T := \frac{Z}{\sqrt{V/\nu}}$

```
double num = norm_rand();
return num / sqrt(rchisq(df) / df);
```

Fun Fact: Cauchy is $t(1)$!

```
location + scale * tan(M_PI * unif_rand());
```

and compute $T := Z/\sqrt{V/\nu}$

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Goodness of Fit testing

Lets say we have an algorithm for generating $Z_i \sim F(t)$

Does it actually work?

- Hypothesis test

$H_0: Z_1, \dots, Z_n \sim F(t)$ ← The Z's come from $F(t)$

$H_1: Z_1, \dots, Z_n \not\sim F(t)$ ← The Zs come from some other distribution.

- Kolmogorov-Smirnov test
- Anderson-Darling test

} any distribution.

- Shapiro-Wilks test ← just for Normal Dist

```
nn = 1000

## Generate datasets with small sample
pit.data = pit.exp( nn );
exp.data = rexp( nn );

ks.test( pit.data, pexp );
```

← generate n Exp(1) random variables with PIT

← Dist to compare to (H_0)

```
    Asymptotic one-sample Kolmogorov-Smirnov test

data:  pit.data
D = 0.022347, p-value = 0.7001
alternative hypothesis: two-sided
```

← p-value

test stat

- p-value is not significant so don't reject H_0 that $Z \sim Exp(1)$

R Code Syntax

rexp → generate random Exp r.v.
dexp → density or PDF of Exp r.v.
pexp → CDF of Exp r.v.
qexp → Quantile of Exp r.v.

* For CDF, $F(t)$ the Quantile function is $F^{-1}(t)$.

also, rnorm, dnorm, pnorm, qnorm for Normal Dist.

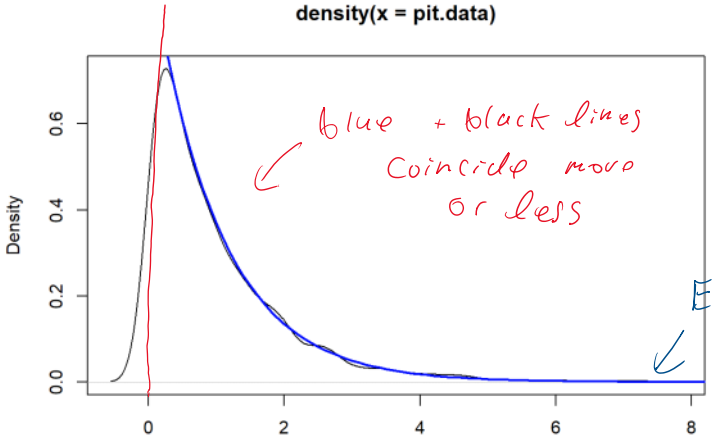
Also Do Goodness of Fit checking by just plotting Data and looking at it!

Also no Goodness of Fit Checking by just plotting Data and looking at it!

```r
nn = 1000

## Generate datasets with small sample

pit.data = pit.exp( nn );
exp.data = rexp( nn );

tt = seq(0,10,0.1)

plot( density( pit.data ) );
lines( tt, dexp(tt), col='blue',lwd=2 );
```

kernel Density Est from the Data

pdf of Exp(1)



blue + black lines coincide more or less

Everything is small

Usually Hypothesis tests only focus on the bulk of the distribution, not the tails / extreme values.